

Core Statistics Topics for Data Science Prerequisites

1. Descriptive Statistics

- Mean, Median, Mode
 - Range, Variance, Standard Deviation
 - Percentiles, Quartiles
 - Skewness and Kurtosis
 - Frequency distributions
 - Data visualization (histograms, boxplots, scatter plots)
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2. Probability Basics

- Sample space, events, outcomes
 - Types of events (independent, mutually exclusive)
 - Conditional probability
 - Bayes' Theorem
 - Permutations and combinations
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3. Probability Distributions

- Discrete Distributions:
 - Bernoulli
 - Binomial
 - Poisson
 - Continuous Distributions:
 - Uniform
 - Normal (Gaussian)
 - Exponential
 - Central Limit Theorem
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4. Statistical Inference

- Population vs. sample
 - Sampling methods and bias
 - Point estimation and interval estimation
 - Confidence intervals
 - Hypothesis testing:
 - Null and alternate hypotheses
 - Type I and Type II errors
 - p-values and significance level
 - One-tailed vs. two-tailed tests
 - t-tests, z-tests, chi-square tests, ANOVA
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5. Correlation and Regression

- Covariance vs. correlation
 - Pearson and Spearman correlation
 - Simple linear regression
 - Multiple linear regression
 - Assumptions of linear regression
 - Interpretation of regression coefficients
 - R-squared and adjusted R-squared
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6. Exploratory Data Analysis (EDA)

- Outlier detection
- Missing value treatment
- Data distributions and transformations
- Feature scaling (Normalization, Standardization)